Notes on Stochastic Dynamics (Spring 2026)

MATH 4900-05

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April 2026

1 Introduction

This note aims to build a foundational pathway toward understanding key results in stochastic dynamics, including contraction mappings, infinitesimal generators, and existence and uniqueness theorems for ODEs and SDEs.

The core philosophy is that advanced results rely heavily on a small set of fundamental analytical tools. Therefore, before approaching these topics, we first establish the necessary background in analysis and metric space theory.

Foundational Concepts (Section A)

We begin with the following essential concepts:

1. **Lipschitz continuity:** including the Lipschitz condition, Lipschitz inequality, and the distinction between global and local Lipschitz continuity. A central question is: why is Lipschitz continuity important?
2. **Function spaces:** including C^2 functions and L^p spaces.
3. **Metric spaces:** definitions and properties such as non-negativity, symmetry, and the triangle inequality.
4. **Cauchy sequences:** understanding convergence from an intrinsic perspective.
5. **Completeness and adaptedness:** completeness in metric spaces and the notion of adapted processes in measure-theoretic probability.

Core Topics (Section B)

With these tools in place, we proceed to study:

1. Contraction mappings
2. Banach fixed point theorem
3. Infinitesimal generators of stochastic processes
4. Existence and uniqueness of ODEs
5. Grönwall inequality
6. Existence and uniqueness of SDEs

Goal

The primary goal of this note is to understand Topics 1–5 in Section B. However, this requires a solid grasp of all concepts listed in Section A.

The overall structure reflects a fundamental principle in analysis:

advanced results are built upon carefully layered foundational tools.

Part I
Section A

2 A1. Lipschitz Continuity

2.1 Definition (Lipschitz Continuity)

Let (X, d_X) and (Y, d_Y) be metric spaces. A function $f : X \to Y$ is called **Lipschitz continuous** if there exists a constant $L > 0$ such that for all $x, y \in X$,

$$d_Y(f(x), f(y)) \leq L \cdot d_X(x, y).$$

The constant L is called a **Lipschitz constant**.

2.2 Special Case: $\mathbb{R}^n \to \mathbb{R}^m$

If $f : \mathbb{R}^n \to \mathbb{R}^m$, then

$$\|f(x) - f(y)\| \leq L\|x - y\|.$$

2.3 Global vs Local Lipschitz

- **Global Lipschitz:** The inequality holds for all $x, y \in X$ with a single constant L.
- **Local Lipschitz:** For every point $x_0 \in X$, there exists a neighborhood U of x_0 and a constant L_U such that

$$\|f(x) - f(y)\| \leq L_U\|x - y\|, \quad \forall x, y \in U.$$

2.4 Lipschitz Implies Continuity

If f is Lipschitz, then f is uniformly continuous, hence continuous.

2.5 Lipschitz Inequality

The defining inequality:

$$|f(x) - f(y)| \leq L|x - y|$$

controls how fast the function can change.

2.6 Why Lipschitz is Important

- Guarantees **uniqueness** in ODE/SDE (Picard iteration).
- Enables **contraction mapping** when $L < 1$.
- Prevents functions from oscillating too wildly.

2.7 Key Example

- $f(x) = x^2$ is **not globally Lipschitz** on $\mathbb{R}$.
- But it is **locally Lipschitz**.

2.8 Derivative Criterion

If $f : \mathbb{R} \to \mathbb{R}$ is differentiable and

$$|f'(x)| \leq M \quad \forall x,$$

then f is Lipschitz with constant $L = M$.

2.9 Important Insight (Core Idea)

Lipschitz condition means:

$$\text{distance in output} \leq \text{constant} \times \text{distance in input}$$

It is a **linear control on growth**.

3 A2. C^k Functions and L^p Spaces

3.1 Definition (C^k Function)

Let $U \subset \mathbb{R}^n$ be an open set. A function $f : U \to \mathbb{R}^m$ is called a C^k **function** if all partial derivatives of f up to order k exist and are continuous.

3.2 Definition (C^2 Function)

A function f is called C^2 if:

- All first-order partial derivatives exist and are continuous
- All second-order partial derivatives exist and are continuous

3.3 Notation

$$f \in C^2(U)$$

3.4 Why C^2 Matters

- Required for **Itô formula** in stochastic calculus
- Ensures second-order Taylor expansion
- Guarantees smoothness needed for PDE/SDE analysis

3.5 Key Insight

$$C^2 \Rightarrow \text{function is smooth enough to do calculus twice}$$

3.6 Example

- $f(x) = x^2 \in C^\infty$
- $f(x) = |x| \notin C^1$

3.7 Definition (L^p Space)

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. For $1 \leq p < \infty$, define:

$$L^p(\Omega) = \left\{ f : \Omega \to \mathbb{R} \;\middle|\; \int_\Omega |f(x)|^p d\mu < \infty \right\}$$

3.8 Norm in L^p

$$\|f\|_p = \left(\int_\Omega |f(x)|^p d\mu \right)^{1/p}$$

3.9 Special Case: L^2

- Most important in stochastic analysis
- Has inner product:

$$\langle f, g \rangle = \int fg \, d\mu$$

3.10 Why L^p Matters

- Measures **size of functions** (not pointwise, but averaged)
- Used to define spaces of random variables
- L^2 is the natural space for Brownian motion and Itô calculus

3.11 Key Insight

$$L^p = \text{functions whose } p\text{-th power is integrable}$$

3.12 Important Distinction

- Pointwise bounded $\neq L^p$
- L^p focuses on **integral control**, not max value

4 A3. Metric Space

4.1 Definition (Metric Space)

A **metric space** is a pair (X, d) where:

- X is a set
- $d : X \times X \to \mathbb{R}$ is a function (called a metric)

such that for all $x, y, z \in X$:

- (Non-negativity)
$$d(x, y) \geq 0$$
- (Identity of indiscernibles)
$$d(x, y) = 0 \iff x = y$$
- (Symmetry)
$$d(x, y) = d(y, x)$$
- (Triangle inequality)
$$d(x, z) \leq d(x, y) + d(y, z)$$

4.2 Key Example

- Euclidean space: $X = \mathbb{R}^n$
$$d(x, y) = \|x - y\|$$

4.3 Why Metric Matters

- Defines **distance** in abstract spaces
- Allows us to define convergence, continuity, and limits
- Foundation for contraction mapping theorem

4.4 Open Ball

For $x \in X$, radius $r > 0$:

$$B(x, r) = \{y \in X : d(x, y) < r\}$$

4.5 Key Insight

Metric = structure that lets us talk about "closeness"

4.6 Important Example (Function Space)

Let $X = C([0,T])$ (continuous functions). Define:

$$d(f,g) = \sup_{t\in[0,T]} |f(t) - g(t)|$$

This is crucial for ODE/SDE (Picard iteration lives here).

5 A4. Cauchy Sequence and Completeness

5.1 Definition (Cauchy Sequence)

Let (X,d) be a metric space. A sequence $\{x_n\}$ is called a **Cauchy sequence** if:

$$\forall \varepsilon > 0,\ \exists N \in \mathbb{N} \text{ such that } m,n \geq N \Rightarrow d(x_m, x_n) < \varepsilon.$$

5.2 Key Insight

All terms of the sequence get arbitrarily close to each other

5.3 Important Idea

A Cauchy sequence does **not** need to know its limit — it only needs to internally converge.

5.4 Definition (Convergent Sequence)

A sequence $\{x_n\}$ converges to $x \in X$ if:

$$d(x_n, x) \to 0.$$

5.5 Relation

- Convergent $\Rightarrow$ Cauchy (always true)
- Cauchy $\Rightarrow$ Convergent (not always true!)

5.6 Definition (Complete Metric Space)

A metric space (X,d) is called **complete** if:

every Cauchy sequence converges to a point in X.

5.7 Key Insight

$$\text{Complete} = \text{no "missing limit points"}$$

5.8 Example

- $\mathbb{R}$ is complete
- $\mathbb{Q}$ is not complete (e.g., sequence converging to $\sqrt{2}$)

5.9 Why Completeness Matters

- Needed for Banach Fixed Point Theorem
- Guarantees limits exist inside the space
- Ensures iterative methods converge properly

5.10 Important Function Space

Let $X = C([0, T])$ with metric:

$$d(f, g) = \sup_{t \in [0,T]} |f(t) - g(t)|.$$

Then (X, d) is **complete**.

6 A5. Completeness and Adapted Processes

6.1 Recap: Complete Metric Space

A metric space (X, d) is **complete** if every Cauchy sequence converges to a limit in X.

6.2 Key Insight

$$\text{No limit points are missing}$$

6.3 Probability Setup

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.

6.4 Filtration

A **filtration** $\{\mathcal{F}_t\}_{t \geq 0}$ is an increasing family of σ-algebras:

$$\mathcal{F}_s \subseteq \mathcal{F}_t \quad \text{for } s \leq t.$$

6.5 Interpretation

$$\mathcal{F}_t = \text{all information available up to time } t$$

6.6 Definition (Adapted Process)

A stochastic process $\{X_t\}_{t\geq 0}$ is called **adapted** to $\{\mathcal{F}_t\}$ if:

$$X_t \text{ is } \mathcal{F}_t\text{-measurable for all } t.$$

6.7 Key Insight

$$\text{At time } t,\ X_t \text{ only depends on information up to } t$$

6.8 Why Adapted Matters

- Prevents using **future information**
- Essential for defining Itô integral
- Ensures models are physically/financially meaningful

6.9 Example

- Brownian motion B_t is adapted to its natural filtration
- $X_t = B_{t+1}$ is **not adapted**

Part II
Section B

7 B1,B2. Contraction Mapping and Banach Fixed Point Theorem

7.1 Definition (Contraction Mapping)

Let (M, d) be a metric space. A mapping $T : M \to M$ is called a **contraction** if there exists $\alpha \in (0, 1)$ such that

$$d(T(x), T(y)) \leq \alpha d(x, y), \quad \forall x, y \in M.$$

7.2 Key Insight

$$\text{The map strictly shrinks distances}$$

7.3 Relation to Lipschitz

A contraction is a Lipschitz continuous function with Lipschitz constant $\alpha < 1$.

7.4 Definition (Fixed Point)

A point $x^* \in M$ is called a **fixed point** if

$$T(x^*) = x^*.$$

7.5 Theorem (Banach Fixed Point Theorem)

Let (M, d) be a **complete** metric space and $T : M \to M$ a contraction. Then:

- There exists a unique fixed point $x^* \in M$
- For any $x_0 \in M$, define $x_{n+1} = T(x_n)$, then

$$x_n \to x^*$$

7.6 Proof Sketch (Core Idea)

Start with $x_{n+1} = T(x_n)$.

- Step 1: Show contraction:

$$d(x_{n+1}, x_n) \leq \alpha^n d(x_1, x_0)$$

- Step 2: Show Cauchy:

$$d(x_n, x_m) \leq \sum_{k=n}^{m-1} \alpha^k d(x_1, x_0)$$

- Step 3: Since $\alpha < 1$, geometric series converges
- Step 4: Completeness $\Rightarrow$ limit exists
- Step 5: Take limit:

$$x^* = T(x^*)$$

- Step 6: Uniqueness:

$$d(x^*, y^*) \leq \alpha d(x^*, y^*) \Rightarrow x^* = y^*$$

8 B3. Infinitesimal Generator

Let $\{X_t\}_{t \geq 0}$ be a stochastic process. The **infinitesimal generator** L is defined by:

$$Lf(x) = \lim_{t \to 0} \frac{\mathbb{E}[f(X_t) \mid X_0 = x] - f(x)}{t}$$

8.1 Key Insight

Generator describes instantaneous evolution of expectations

8.2 Example (Brownian Motion)

For Brownian motion B_t:

$$Lf(x) = \frac{1}{2}f''(x)$$

8.3 Why Generator Matters

- Connects stochastic processes to PDEs
- Leads to Kolmogorov equations
- Fundamental in stochastic analysis and diffusion theory

9 B4. Existence and Uniqueness of ODEs

9.1 The Initial Value Problem

Consider the ODE

$$\begin{cases} x'(t) = b(t, x(t)), \\ x(0) = x_0. \end{cases}$$

We want to know whether there exists a solution, and whether that solution is unique.

9.2 Integral Form

A function $x(t)$ solves the ODE if and only if it satisfies

$$x(t) = x_0 + \int_0^t b(s, x(s))\, ds.$$

This suggests defining an operator T by

$$(Tx)(t) := x_0 + \int_0^t b(s, x(s))\, ds.$$

Then solving the ODE is equivalent to finding a fixed point of T:

$$Tx = x.$$

9.3 Assumptions

Assume:

- b is continuous in t
- b is Lipschitz continuous in x, i.e. there exists $L > 0$ such that

$$|b(t,x) - b(t,y)| \leq L|x-y|$$

  for all t, x, y

9.4 Function Space

Let

$$X = C([0,T], \mathbb{R}^d)$$

with metric

$$d(x,y) = \sup_{t\in[0,T]} |x(t) - y(t)|.$$

This is a complete metric space.

9.5 Why T is a Contraction for Small T

For $x, y \in X$,

$$\begin{aligned}
|(Tx)(t) - (Ty)(t)| &= \left| \int_0^t \big(b(s,x(s)) - b(s,y(s))\big)\, ds \right| \\
&\leq \int_0^t |b(s,x(s)) - b(s,y(s))|\, ds \\
&\leq \int_0^t L|x(s) - y(s)|\, ds \\
&\leq LT \sup_{s\in[0,T]} |x(s) - y(s)|.
\end{aligned}$$

Therefore,

$$d(Tx, Ty) \leq LT\, d(x,y).$$

So if $LT < 1$, then T is a contraction.

9.6 Local Existence and Uniqueness

If $LT < 1$, then by Banach Fixed Point Theorem, T has a unique fixed point in X.

Hence the ODE has a unique solution on $[0,T]$.

9.7 Key Idea

$$\text{ODE} \quad \Longrightarrow \quad \text{integral equation} \quad \Longrightarrow \quad \text{fixed point problem}$$

9.8 Why Lipschitz Matters

The Lipschitz condition is what gives uniqueness. Without Lipschitz continuity, solutions may fail to be unique.

9.9 Why Small Time Matters

The factor LT appears in the contraction estimate. Even if L is fixed, we can choose T small enough so that $LT < 1$.

9.10 Conclusion

Under continuity in t and Lipschitz continuity in x, the initial value problem

$$x'(t) = b(t, x(t)), \qquad x(0) = x_0$$

has a unique local solution.

10 B5. Gronwall Inequality

10.1 Basic Form

Let $u(t)$ be a nonnegative function satisfying

$$u(t) \leq C + \int_0^t K\, u(s)\, ds$$

for constants $C \geq 0$, $K \geq 0$. Then

$$u(t) \leq Ce^{Kt}.$$

10.2 Differential Form

If $u(t)$ satisfies

$$u'(t) \leq Ku(t),$$

then

$$u(t) \leq u(0)e^{Kt}.$$

10.3 Key Insight

$$\text{Integral inequality} \;\Rightarrow\; \text{exponential bound}$$

10.4 Why It Matters

- Controls growth of solutions
- Used to prove uniqueness of ODE/SDE
- Bounds the difference between two solutions

10.5 Typical Use (Difference of Two Solutions)

Let $x(t), y(t)$ be two solutions. Define

$$u(t) = |x(t) - y(t)|.$$

Then often we get

$$u(t) \leq \int_0^t Lu(s)\, ds.$$

By Gronwall,

$$u(t) = 0 \Rightarrow x(t) = y(t).$$

10.6 Key Conclusion

If initial data is the same, solutions must coincide

11 B6. Existence and Uniqueness of SDEs

11.1 SDE (Itô Form)

Consider

$$X_t = X_0 + \int_0^t b(s, X_s)\, ds + \int_0^t \sigma(s, X_s)\, dB_s,$$

where B_t is Brownian motion.

11.2 Assumptions

Assume for some $L, K > 0$ and all t, x, y:

- (Lipschitz)

$$|b(t,x) - b(t,y)| + \|\sigma(t,x) - \sigma(t,y)\| \leq L|x - y|$$

- (Linear growth)

$$|b(t,x)|^2 + \|\sigma(t,x)\|^2 \leq K(1 + |x|^2)$$

11.3 Function Space

Work in $L^2(\Omega; C([0,T]; \mathbb{R}^d))$ with metric

$$d(X,Y) := \left(\mathbb{E}\big[\sup_{t \in [0,T]} |X_t - Y_t|^2 \big] \right)^{1/2}.$$

11.4 Define the Operator

For a process X, define

$$(TX)_t := X_0 + \int_0^t b(s, X_s)\, ds + \int_0^t \sigma(s, X_s)\, dB_s.$$

11.5 Estimate (Sketch)

For X, Y,

$$\mathbb{E} \sup_{t \le T} |(TX)_t - (TY)_t|^2 \le C \int_0^T \mathbb{E} \sup_{r \le s} |X_r - Y_r|^2\, ds,$$

using Lipschitz + Itô isometry / BDG inequality.

11.6 Apply Gronwall

Let

$$u(t) = \mathbb{E} \sup_{s \le t} |X_s - Y_s|^2.$$

Then

$$u(t) \le C \int_0^t u(s)\, ds.$$

By Gronwall,

$$u(t) = 0 \Rightarrow X = Y.$$

11.7 Conclusion

Under Lipschitz and linear growth conditions, the SDE has a unique solution in L^2.